30. Phenols undergo electrophilic substitution reactions readily due to the strong activating effect of OH group attached to the benzene ring. Since, the OH group increases the electron density more to o– and p– positions therefore OH group is ortho, para-directing. Reimer-Tiemann reaction is one of the examples of aldehyde group being introduced on the aromatic ring of phenol, ortho to the hydroxyl group. This is a general method used for the ortho-formylation of phenols.

Answer the following questions :

(a) What happens when phenol reacts with **2**

(i) Br_2/CS_2

(ii) Conc. HNO_3

(b) Why phenol does not undergo protonation readily ? **1**

(c) Which is a stronger acid – phenol or cresol ? Give reason. **1**

OR

(c) Write the IUPAC name of the product formed in the Reimer-Tiemann reaction. **1**

SECTION – B

17. (A) Give reasons : **(1 + 1 = 2)**

(a) Cooking is faster in pressure cooker than in an open pan.

(b) On mixing liquid X and liquid Y, volume of the resulting solution decreases. What type of deviation from Raoult's law is shown by the resulting solution ? What change in temperature would you observe after mixing liquids X and Y ?

OR

(B) Define Azeotrope. What type of Azeotrope is formed by negative deviation from Raoult's law ? Give an example. **2**

23. (a) State the following : **3**

(i) Kohlrausch law of independent migration of ions and

(ii) Faraday's first law of electrolysis.

(b) Using E°_{values} of X and Y given below, predict which is better for coating the surface of iron to prevent corrosion and why ?

Given $E^{\circ}_{X^{2+}/X} = -2.36$ V,

$E^{\circ}_{Y^{2+}/Y} = -0.14$ V,

$E^{\circ}_{Fe^{2+}/Fe} = -0.44$ V

24. A certain reaction is 50% complete in 20 minutes at 300 K and the same reaction is 50% complete in 5 minutes at 350 K. Calculate the activation energy if it is a first order reaction. **3**

[$R = 8.314\ J\ K^{-1}\ mol^{-1}$; $\log 4 = 0.602$]

25. The elements of 3d transition series are given as : **(1 + 1 + 1 = 3)**

Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn

Answer the following :

(a) Copper has exceptionally positive $E^{\circ}_{M^{2+}/M}$ value, why ?

(b) Which element is a strong reducing agent in +2 oxidation state and why ?

(c) Zn^{2+} salts are colourless. Why ?